

Nexora Internal Technical Manual v3.2

Section 1 — Company Snapshot

Nexora Technologies closed FY2024 with **\$512 million** in revenue and a global headcount of **3,150 employees**. (Note: these are the audited internal figures and differ from the preliminary numbers in the external annual report.)

Section 2 — Retrieval-Augmented Generation (RAG)

RAG combines a retriever (which fetches relevant documents) with a generator (which produces an answer conditioned on those documents). Standard RAG retrieves once and generates once; it has no mechanism to detect whether retrieval succeeded.

2.1 HyDE — Hypothetical Document Embeddings

HyDE asks the LLM to first generate a HYPOTHETICAL answer to the query, then uses that hypothetical document as the search query. Because answer-to-answer semantic matching is more precise than question-to-answer matching, HyDE improves retrieval recall on abstract questions. The hypothetical document is discarded after retrieval — only the real retrieved documents reach the generator.

2.2 Cross-Encoder Reranking

A cross-encoder scores (query, document) pairs jointly using full attention across both texts. This is far more accurate than bi-encoder similarity (used for initial vector search) but slower, because every (query, doc) pair requires a separate forward pass. The standard pattern is two-stage: a fast bi-encoder retrieves candidates, then a cross-encoder reranks the top-K.

2.3 Parent-Child Chunking

Parent-child chunking splits each document into large PARENT chunks (e.g. 1024 tokens) and small CHILD chunks (e.g. 256 tokens). Children are embedded and retrieved for precise matching; when a child matches, the system returns the full PARENT as generation context. This gives the LLM coherent, broader context than the tiny retrieved chunk alone.

2.4 BM25 vs Vector Search

BM25 is a lexical (keyword) retriever: it scores documents by term frequency and inverse document frequency. It excels when queries share exact vocabulary with documents (names, IDs, code, jargon). Vector search embeds both query and documents into a semantic space and matches by cosine similarity; it excels when meaning matters more than wording. For keyword-heavy queries — exact product names, error codes, technical identifiers — BM25 typically outperforms vector search. Hybrid retrieval fuses both via Reciprocal Rank Fusion (RRF) to get the strengths of each.

2.5 CRAG — Corrective RAG

CRAG adds a self-correction layer to RAG. After retrieval, each document is GRADED for relevance. If the retrieval is judged poor, CRAG falls back to web search before generating. Standard RAG has no

such check — it generates from whatever was retrieved, even if irrelevant. Use CRAG when retrieval quality is uncertain or the corpus is noisy; use plain RAG when the corpus is clean and queries reliably match.

Section 3 — Formatting Notes (intentionally messy)

This manual mixes font sizes, irregular spacing, and inconsistent heading levels to stress-test the document loader.